

Assignment 2 — Full-Stack Web Application

Group project | Weight: **20%** | Node.js · Express · MongoDB · Pug

Task

In a group of 3–4, build a database-driven web app using Node.js/Express, MongoDB, and the **Pug** template engine. The app must perform full **CRUD** (Create, Read, Update, Delete) on a primary resource. **Choose your own topic** — e.g. a recipe manager, book catalogue, event tracker, or task board. Implementation quality matters more than originality.

Example Topics (pick one, or propose your own)

- Recipe manager — recipes with ingredients, category, prep time.
- Book / movie catalogue — title, author/director, genre, rating, review.
- Event tracker — events with date, location, capacity, notes.
- Task / project board — tasks with status, priority, assignee, due date.
- Inventory system — products with SKU, quantity, price, supplier.
- Travel journal — trips with destination, dates, photos link, highlights.
- Fitness / workout log — sessions with exercise, sets, reps, date.
- Contact / CRM directory — contacts with company, email, phone, tags.

Requirements

- **Back end:** Node.js + Express, organised routes/controllers, environment config for secrets.
- **Database:** MongoDB (Mongoose recommended); a defined schema with 5–6 meaningful fields; data persists.
- **Views:** all HTML rendered with Pug — a shared layout plus partials, driven by server data.
- **CRUD:** Create (form + validation), Read (list + detail views), Update (pre-filled edit form), Delete (with confirm).
- **Quality:** server-side validation, error handling (no crashes), presentable UI, readable code.

Timeline

Milestone	Due	Week
Group + proposal (topic, resource, fields)	Roster + 1-paragraph proposal for sign-off	11
Data + skeleton	Repo, DB connected, model, Pug layout/nav	12
CRUD complete	All four operations working end-to-end	13
Final + demo	Polished app, README, presentation	14

Confirm exact due dates on Brightspace.

Deliverables

- Git repo or zip (exclude node_modules) with a valid package.json.
- **README.md**: description, team members, setup/run steps, seed data if any.
- 5–8 min in-class live demo — every member speaks, walk through each CRUD operation.
- Submit via Brightspace by the deadline. One submission per group.

Marks (100 = 20% of final grade)

Category	Marks
Back end (Node.js / Express)	15
Database (MongoDB)	15
Pug templating	15
Full CRUD functionality	25
Validation & error handling	8
UI/UX & code quality	7
Presentation & demo	10
Documentation (README)	5
TOTAL	100

Presentation (10): clear live demo of all CRUD operations, every member contributes, questions handled well.

All work must be your group's own; cite any adapted code. Academic Integrity policy applies.